CEC 315 – Signals and Systems

Exam: Lectures 9–15 (Exam 2)

Rogelio Gracia Otalvaro Spring 2026

Name: _____ **Date:** _____

Instructions:

- Total time: **60 minutes**. Total points: **100**.
- Show all work clearly. Box your final answers.
- Make sure your answers are clear and easy to understand.
- Read the text carefully and understand the questions, there is plenty of time.
- Assume all systems are causal and stable unless stated otherwise.

Useful Formulas:

Euler: $e^{j\theta} = \cos\theta + j\sin\theta, \quad \cos\theta = \dfrac{e^{j\theta}+e^{-j\theta}}{2}, \quad \sin\theta = \dfrac{e^{j\theta}-e^{-j\theta}}{2j}$

CT FS: $a_k = \dfrac{1}{T}\displaystyle\int_T x(t)\,e^{-jk\omega_0 t}\,dt, \quad x(t) = \sum_{k=-\infty}^{\infty} a_k\,e^{jk\omega_0 t}, \quad \omega_0 = \dfrac{2\pi}{T}$

CT FT: $X(j\omega) = \displaystyle\int_{-\infty}^{\infty} x(t)\,e^{-j\omega t}\,dt, \quad x(t) = \dfrac{1}{2\pi}\int_{-\infty}^{\infty} X(j\omega)\,e^{j\omega t}\,d\omega$

FT pair: $e^{-at}u(t) \overset{F}{\longleftrightarrow} \dfrac{1}{a+j\omega}\ (a>0) \qquad \displaystyle\int_0^{\infty} e^{-\alpha t}\,dt = \dfrac{1}{\alpha}, \ \mathrm{Re}(\alpha)>0$

Convolution: $y(t) = x(t) * h(t) \ \Leftrightarrow\ Y(j\omega) = X(j\omega)\cdot H(j\omega)$

Parseval (FS): $\dfrac{1}{T}\displaystyle\int_T |x(t)|^2\,dt = \sum_{k=-\infty}^{\infty} |a_k|^2$

2nd-order: $H(j\omega) = \dfrac{\omega_n^2}{(j\omega)^2 + 2\zeta\omega_n(j\omega) + \omega_n^2}, \quad \%\mathrm{OS} = 100\exp\left(\dfrac{-\pi\zeta}{\sqrt{1-\zeta^2}}\right), \quad t_s \approx \dfrac{4}{\zeta\omega_n}$

Part I: Multiple Choice (5 questions × 4 points = 20 points)

Circle the single best answer for each question.

1. A periodic CT signal $x(t)$ has a jump discontinuity of height A. When the Fourier series is truncated to N harmonics, the peak overshoot near the discontinuity is approximately:
(a) 0% of A (the overshoot vanishes as N increases)
(b) 5% of A
(c) 9% of A, regardless of N
(d) 9% of A, but only when $N < 100$

2. A signal $x(t)$ has Fourier transform $X(j\omega)$. The signal $y(t) = x(t) \cdot e^{j\omega_0 t}$ has Fourier transform:
(a) $e^{-j\omega\omega_0}\, X(j\omega)$
(b) $X\big(j(\omega - \omega_0)\big)$
(c) $X\big(j(\omega + \omega_0)\big)$
(d) $\dfrac{1}{2\pi} X(j\omega) * \delta(\omega - \omega_0)$

3. A discrete-time periodic signal has period $N = 8$. How many distinct Fourier series coefficients a_k does it have, and why?
(a) Infinitely many, because the synthesis sum extends from $-\infty$ to $+\infty$
(b) 8, because DT complex exponentials that differ in frequency by 2π are identical
(c) 8, because the Dirichlet conditions limit the number of harmonics
(d) 16, because both positive and negative frequencies must be counted separately

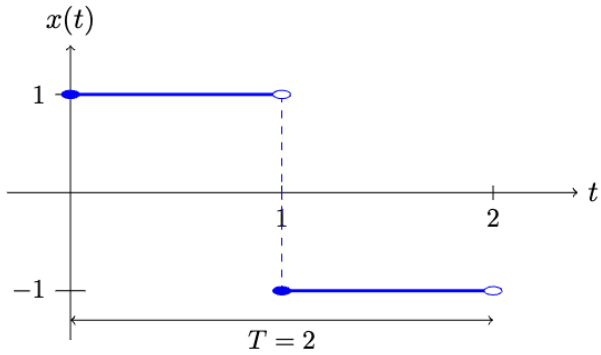
4. A first-order lowpass system has frequency response $H(j\omega) = \dfrac{1}{1 + j\omega/\omega_c}$. On a Bode magnitude plot, the high-frequency asymptotic slope is:
(a) -40 dB/decade
(b) -20 dB/decade
(c) 0 dB/decade
(d) $+20$ dB/decade

5. A second-order CT system has natural frequency $\omega_n = 10$ rad/s and damping ratio $\zeta = 0.3$. Which of the following is true?
(a) The system is overdamped with no resonance peak.
(b) The system is underdamped, and a resonance peak exists because $\zeta < 1/\sqrt{2}$.
(c) The system is underdamped, but no resonance peak exists because $\zeta > 1/\sqrt{2}$.
(d) The system is critically damped.

Part II: Problems (4 problems = 80 points)

Problem 1: CT Fourier Series and Properties (20 points)

A real-valued periodic signal $x(t)$ has period $T = 2$ (so $\omega_0 = \pi$ rad/s). One full period is shown below:



(a) **(5 pts)** Write the piecewise expression for $x(t)$ over one period. Compute a_0 and explain its physical meaning.

(b) **(8 pts)** Derive the general expression for a_k ($k \neq 0$) using the analysis integral. Simplify as far as possible. *Hint:* Factor $e^{-jk\pi}$ and use $e^{-jk\pi} = (-1)^k$.

(c) **(4 pts)** Evaluate $|a_1|$, $|a_2|$, and $|a_3|$ numerically. How do the magnitudes decay with k? Is this consistent with what you expect for a signal with jump discontinuities?

(d) **(3 pts)** Using Parseval's theorem, write an expression for the average power of $x(t)$ in terms of $\{a_k\}$. Compute the average power directly from $x(t)$ and state what it equals.

Problem 2: Fourier Transform Properties and Convolution (25 points)

Let $x(t) = e^{-3t}u(t)$, so that $X(j\omega) = \dfrac{1}{3 + j\omega}$.

(a) **(5 pts)** *Time-shifting property.* Define $y(t) = e^{-3(t-4)}\,u(t-4)$. Use the time-shifting property to find $Y(j\omega)$. Show that $|Y(j\omega)| = |X(j\omega)|$ and explain why this must be true.

(b) **(5 pts)** *Frequency-shifting property.* Find the Fourier transform of

$$z(t) = e^{-3t}u(t) \cdot \cos(8t)$$

Express $Z(j\omega)$ as a sum of two terms. *Hint:* Write $\cos(8t)$ using Euler's formula and apply the frequency-shifting property to each term.

(c) **(7 pts)** *Convolution property.* An LTI system has impulse response $h(t) = 6e^{-5t}u(t)$. The input is $x(t) = e^{-3t}u(t)$.

(i) Find $Y(j\omega) = X(j\omega) \cdot H(j\omega)$.

(ii) Perform partial fraction expansion to write $Y(j\omega) = \dfrac{A}{3 + j\omega} + \dfrac{B}{5 + j\omega}$.

(iii) Find the constants A and B.

(d) **(4 pts)** Invert each term to find $y(t)$ for $t \geq 0$.

(e) **(4 pts)** Verify your answer by checking $y(0)$ and $\lim_{t\to\infty} y(t)$. Are these values consistent with the convolution integral $y(t) = \int_0^t 6e^{-5\tau} \cdot e^{-3(t-\tau)}\, d\tau$?

Problem 3: Frequency Response and Filtering (20 points)

A CT LTI system has impulse response $h(t) = 4e^{-4t}u(t)$, and the input is:

$$x(t) = 3 + 2\cos(4t) + \cos(20t)$$

(a) **(4 pts)** Compute $H(j\omega)$ and identify the -3 dB cutoff frequency ω_c.

(b) **(8 pts)** Evaluate $H(j\omega)$ at each frequency present in $x(t)$. Fill in the table:

ω (rad/s)	$H(j\omega)$	$\lvert H(j\omega)\rvert$	$\angle H(j\omega)$ (degrees)
0			
4			
20			

(c) **(5 pts)** Write the complete output $y(t)$ in trigonometric form.

(d) **(3 pts)** By how many dB is the $\omega = 20$ component attenuated relative to the $\omega = 4$ component at the output? Does this system behave as a lowpass or highpass filter?

Problem 4: Second-Order System and Bode Plot (15 points)

A second-order CT system has the standard-form frequency response:

$$H(j\omega) = \frac{\omega_n^2}{(j\omega)^2 + 2\zeta\omega_n(j\omega) + \omega_n^2}$$

with $\omega_n = 10$ rad/s and $\zeta = 0.5$.

(a) **(3 pts)** Classify the system as underdamped, critically damped, or overdamped. Does a resonance peak exist in $|H(j\omega)|$? Justify.

(b) **(5 pts)** Compute the following quantities:

- DC gain $|H(0)|$
- $|H(j\omega_n)|$ in dB
- Percent overshoot of the step response: $\%\text{OS} = 100\exp\left(\frac{-\pi\zeta}{\sqrt{1-\zeta^2}}\right)$
- Settling time $t_s \approx 4/(\zeta\omega_n)$

(c) **(4 pts)** Describe the Bode magnitude plot asymptotes:

- What is the low-frequency asymptote (in dB)?
- What is the high-frequency asymptotic slope (in dB/decade)?
- At what frequency do the two asymptotes intersect?

(d) **(3 pts)** What phase does $H(j\omega)$ reach at $\omega = \omega_n$, regardless of ζ? What are the limiting phase values as $\omega \to 0$ and $\omega \to \infty$?

End of Exam